SANJANA REDDY MALLIPEDDI

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PROFESSIONAL SUMMARY

Data Analyst with experience analyzing large-scale datasets using **Python, SQL, and statistical modeling** to generate actionable insights and support data-driven decision-making. Skilled in **machine learning, experimentation, and predictive analytics**, with experience building dashboards, developing analytical datasets, and evaluating models to improve product and operational performance.

EDUCATION

Illinois Institute of Technology
Chicago, IL
Master's in Computer Science **(3.5/4.0 GPA)** — Aug 2023 – May 2025
Relevant Coursework: Software Engineering, Distributed Systems, Cloud Computing, Data Structures & Algorithms;

Indian Institute of Information and Technology — Chennai, India
Bachelor's in Computer Science & Engineering — July 2019 – May 2023

Teaching Assistant for Database Organization course, supporting instruction on relational databases, SQL queries, schema design, and database management concepts for undergraduate students.

TECHNICAL SKILLS

Programming: Python, R, SQL, PySpark, Scala, Java, C++
Statistics & Experimentation: Hypothesis Testing, A/B Testing, Regression Analysis, Descriptive Statistics, Correlation Analysis, Statistical Modeling
Machine Learning: Supervised & Unsupervised Learning, Predictive Modeling, Feature Engineering, Model Validation, Performance Evaluation
Data Visualization: Tableau, Power BI, Matplotlib, Seaborn
Databases & Warehousing: PostgreSQL, MySQL, MongoDB, Snowflake, DynamoDB
Data Engineering: Spark, Kafka, Airflow, Data Ingestion, Data Transformation, ETL Pipelines
Cloud & Platforms: AWS (S3, EC2, RDS, Lambda), Databricks, Docker, Git, Linux
Analytics Tools: SAS, SPSS, Informatica, Talend

PROFESSIONAL EXPERIENCE

Republic Services — Scottsdale, AZ
Data Analyst Intern — April 2025 – Present

- Analyzed large-scale operational datasets using **Python and complex SQL queries**, identifying patterns and trends that improved data reliability and supported analytics initiatives.
- Conducted **exploratory data analysis and statistical hypothesis testing** to evaluate relationships between operational variables and key performance metrics.
- Built and maintained analytical datasets on **AWS S3 and Databricks**, enabling scalable analytics workflows and supporting machine learning model development.
- Assisted in building and validating **machine learning models** to forecast operational outcomes and improve predictive insights for internal analytics teams.
- Designed **Tableau and Power BI dashboards** to track KPIs and operational metrics, enabling stakeholders to monitor performance and identify opportunities for optimization.
- Extracted and aggregated large datasets using SQL to support **product analytics and operational decision-making** across multiple teams.

Solvent Software Solutions
Hyderabad, India

Data Analyst Jan 2022 – May 2023

- Extracted and analyzed structured and semi-structured datasets using **SQL and Python**, improving data quality and enabling more reliable analytics workflows.
- Performed **feature engineering and data transformation** on JSON and Parquet datasets to prepare training data for predictive modeling and statistical analysis.
- Developed automated **data ingestion and validation pipelines** using Python, improving efficiency and scalability of analytical processes.
- Conducted statistical analysis including **correlation and regression analysis** to evaluate relationships between operational metrics and business performance indicators.
- Collaborated with data scientists to validate model inputs and assess model outputs using performance metrics to ensure reliability of machine learning pipelines.
- Generated reports and visualizations to communicate analytical findings and support **data-driven decision-making**.

PROJECTS

Ride Demand Forecasting using Time Series May 2024
Tools: Python, SQL, Pandas, Scikit-learn

- Built **time-series forecasting models** to predict ride demand patterns using historical transportation datasets.
- Performed feature engineering on temporal variables (hour, day, seasonality) and location attributes to improve prediction accuracy.
- Evaluated model performance using **RMSE and MAE** to identify the most reliable forecasting model for demand prediction.
- Visualized demand trends and predictions to highlight peak usage periods and support improved allocation of transportation resources.

Product Experimentation and A/B Testing Analysis.
Dec 2024
Tools: Python, SQL, Pandas, SciPy

- Designed and simulated **A/B experiments** to evaluate the impact of product feature changes on user engagement and conversion metrics.
- Conducted **statistical hypothesis testing**, calculating confidence intervals and p-values to determine experiment significance.
- Analyzed user behavior metrics and defined **KPIs to measure experiment performance**, generating insights for product optimization.
- Developed visualizations to communicate experiment outcomes and support data-driven product decision-making.

Real-Time Data Analysis with Kafka and Spark.
June 2025
Tools: Python, Apache Kafka, Spark Structured Streaming

- Developed a **real-time streaming data pipeline** using Kafka and Spark to process high-volume event data.
- Performed exploratory data analysis to identify **user engagement trends and behavioral patterns** within streaming datasets.
- Designed data transformation workflows to generate curated datasets for dashboards and monitoring tools.
- Enabled near real-time analytics to track key performance indicators and support proactive operational decision-making.